Integrated Flood Hazard and Settlement Vulnerability Assessment in Khagaria District, Bihar Using Geospatial Techniques and the AHP Method

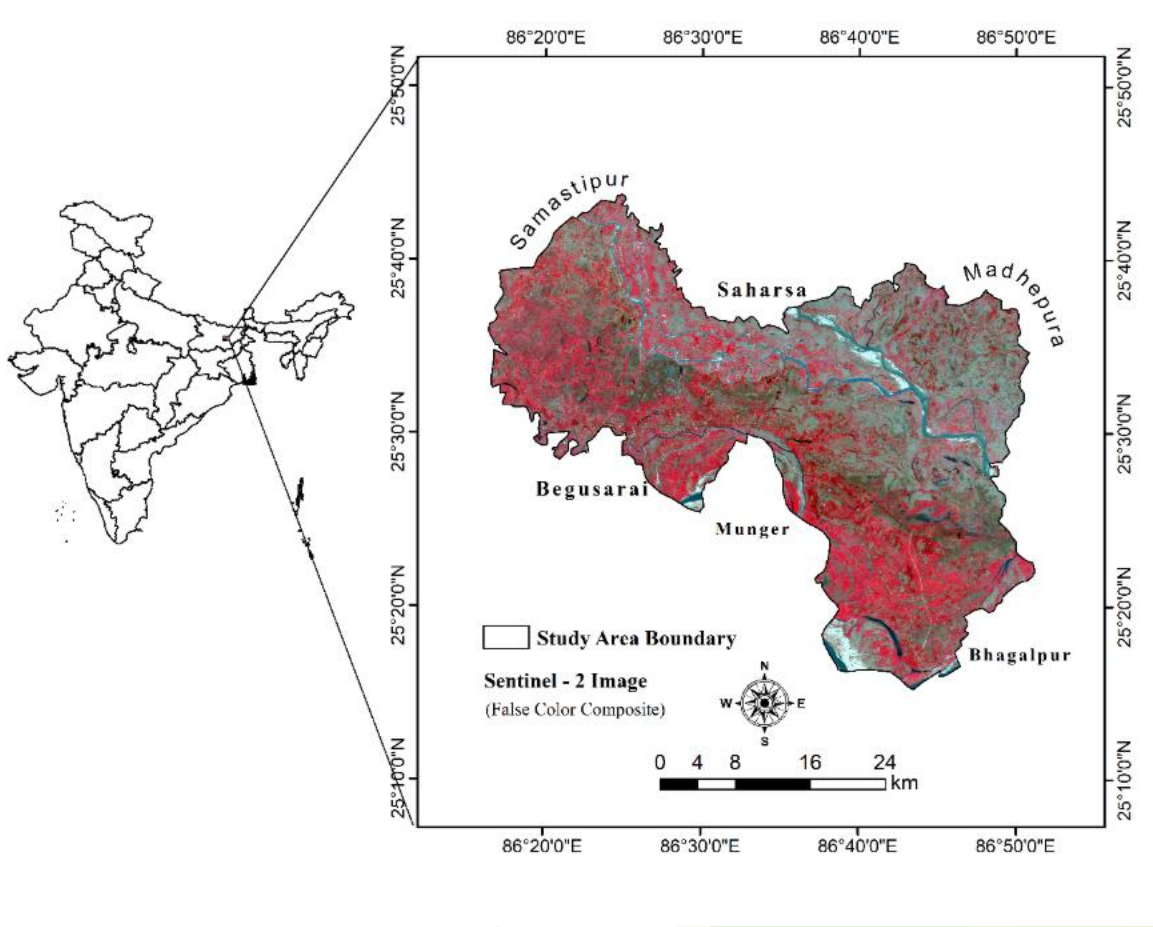
Introduction

The flood history of Khagaria district in Bihar has been catastrophic, with over 74% of the region susceptible to flooding. The significant floods that transpired in 1954, 1957, 1974, 1987, 2004, 2007, 2017, 2020, 2022, and 2024 profoundly affected individuals' lives, properties, and the district's agricultural sector. The current research seeks to employ a multicriteria decision-making framework within a Geographical Information System (GIS), integrating remotely sensed data, to delineate flood-prone areas in Khagaria district. The research employed an 11x11 decision matrix to assess the significance of each parameter in generating the final flood map via the Analytic Hierarchy Process (AHP). The parameters included are elevation, slope, rainfall, drainage density (DD), distance from river (DFR), topographic wetness index (TWI), stream power index (SPI), land use/land cover (LULC), normalised difference vegetation index (NDVI), lithology, and ruggedness index.

Aims, Objective, Study Area and Dataset

Aim and objective - The present study aims to identify flood-prone areas in the Khagaria district and propose appropriate flood mitigation strategies by employing a multicriteria decision-making approach inside a GIS framework, utilising remotely sensed data.

SNO.	DATA TYPES	SOURCES
1.	Rainfall Data	Climate Research Unit (CRU) https://crudata.uea.ac.uk/cru/data/hrg/.
2.	SRTM-DEM (30m resolution)	USGS Earth Explorer website (https://earthexplorer.usgs.gov/).
3.	Sentinal-1 Data (10 m spatial resolution)	Copernicus Open Access Hub https://www.copernicus.eu/en/access-data/conventional-data-access-hubs
4.	Sentinal-2 Data (10 m spatial resolution)	Copernicus Open Access Hub https://www.copernicus.eu/en/access-data/conventional-data-access-hubs
6.	Lithology Data	Geological Survey of India (GSI), https://www.gsi.gov.in/webcenter/portal/OCBIS

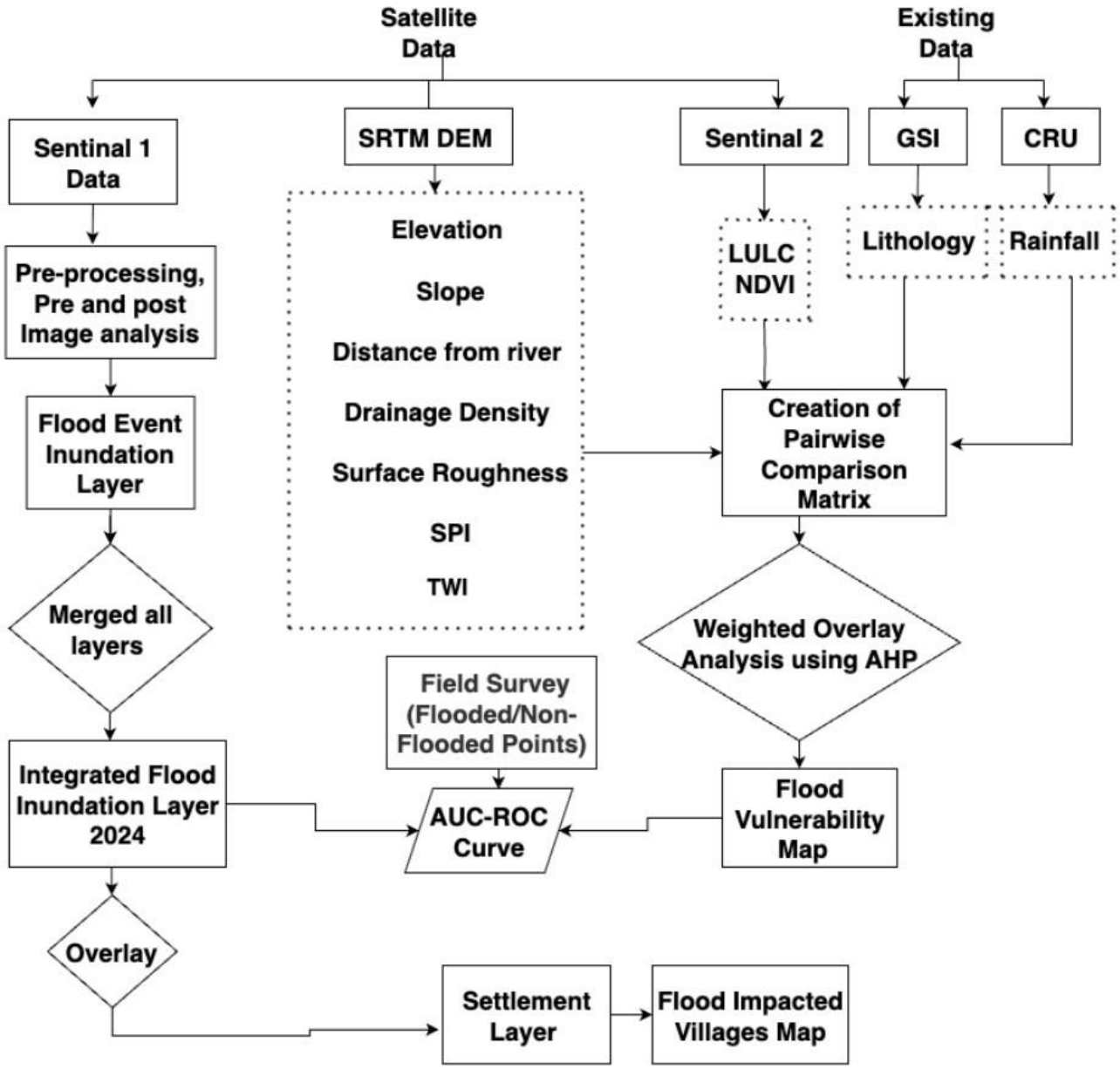


Study Area

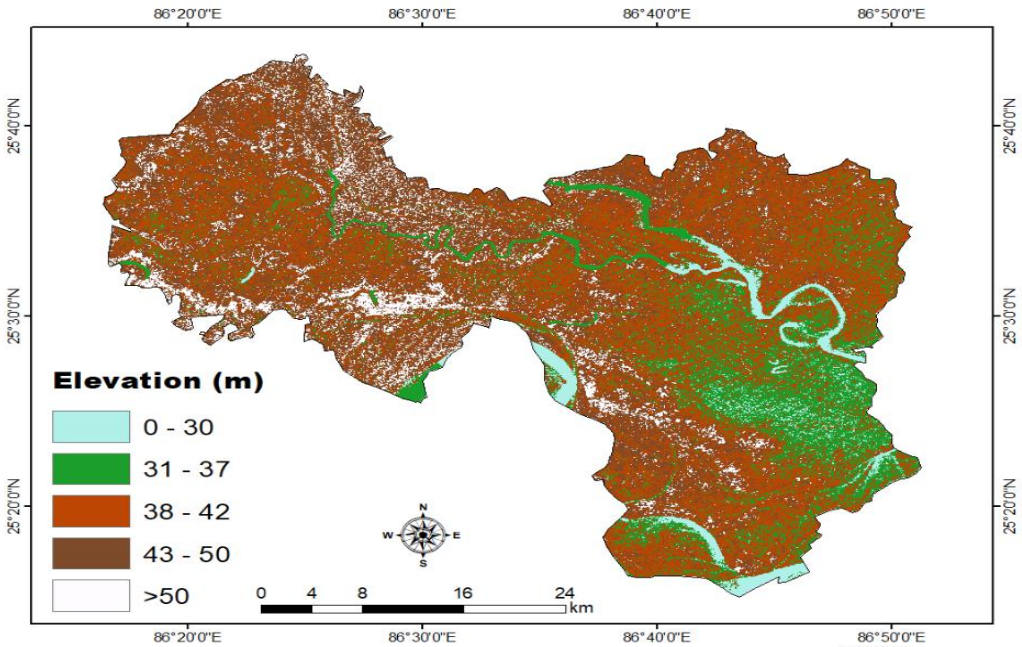
Methodology

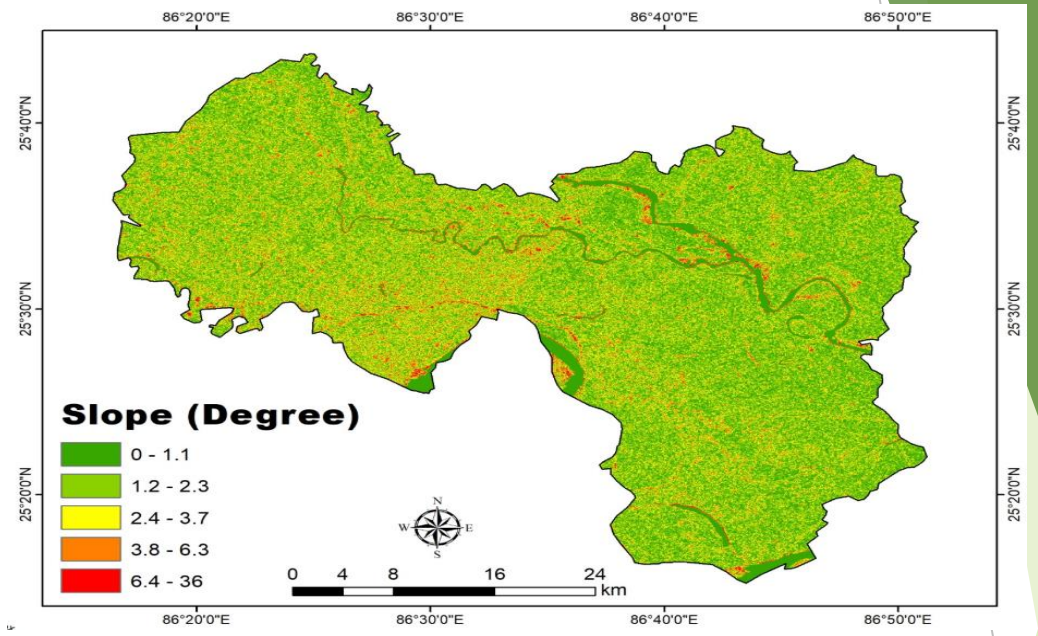
This workflow shows how flood vulnerability is mapped using satellite and existing data. Sentinel and DEM data are processed to derive factors like elevation, slope, drainage density, and land use, while rainfall and lithology are also included.

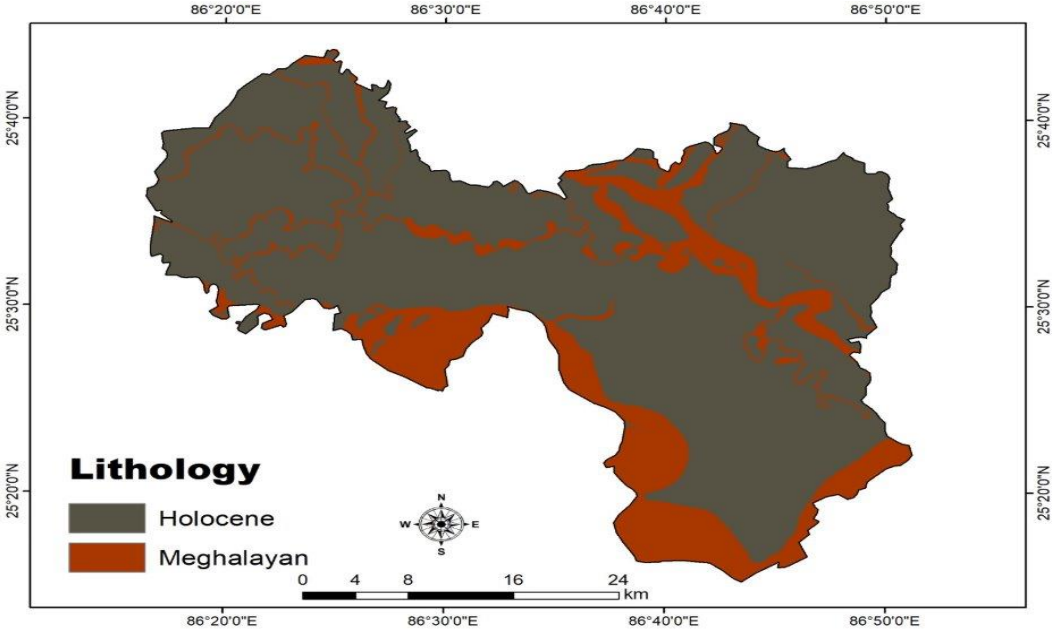
These factors are weighted using AHP (pairwise comparison) and combined through overlay analysis to produce a flood vulnerability map.



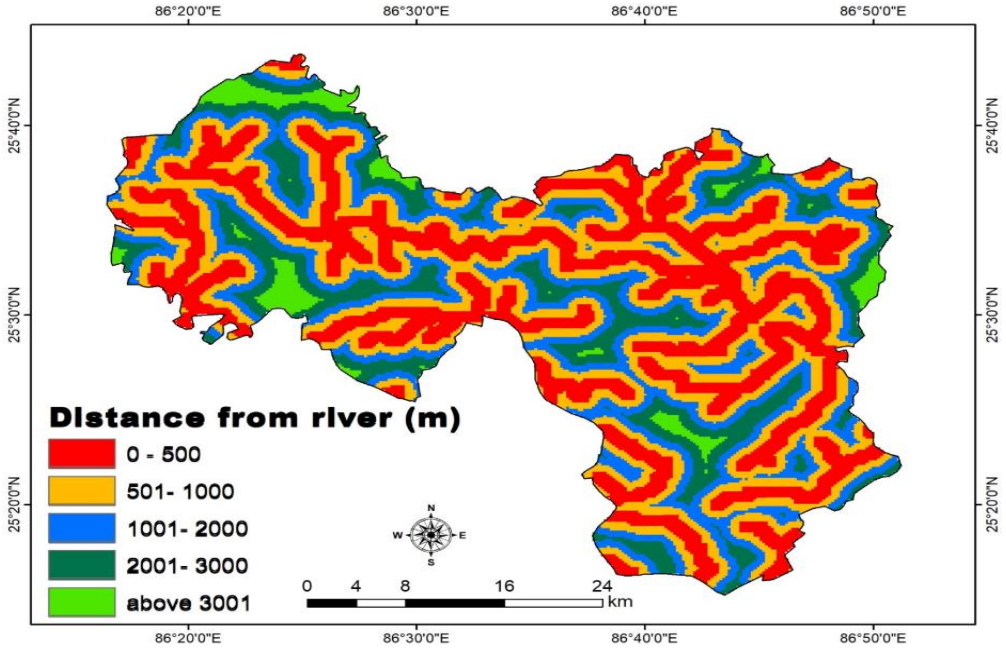
Visual Representation

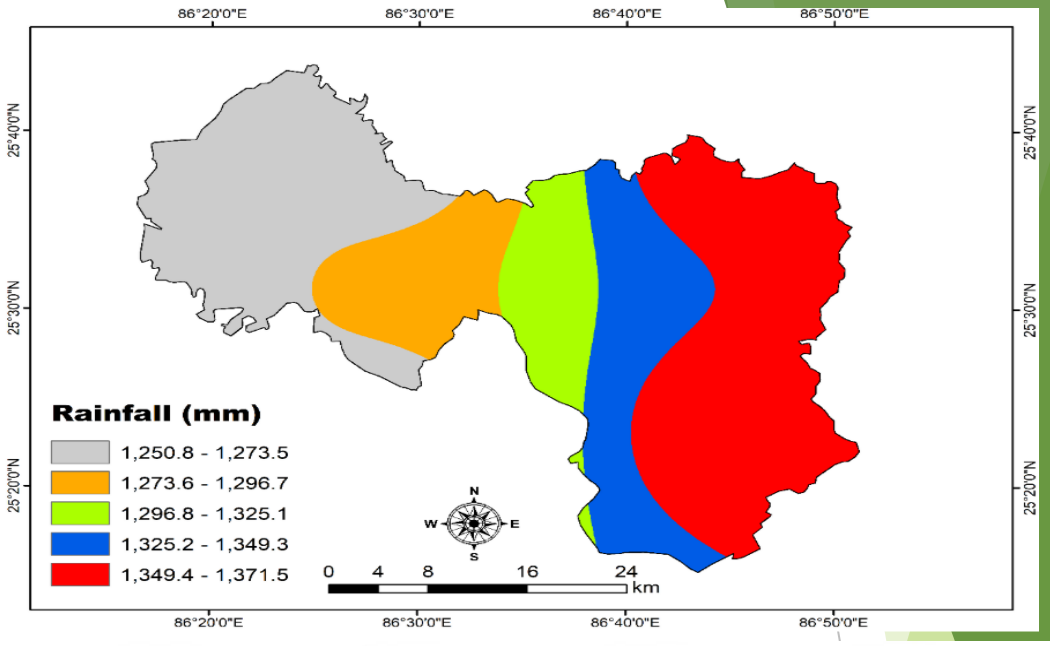


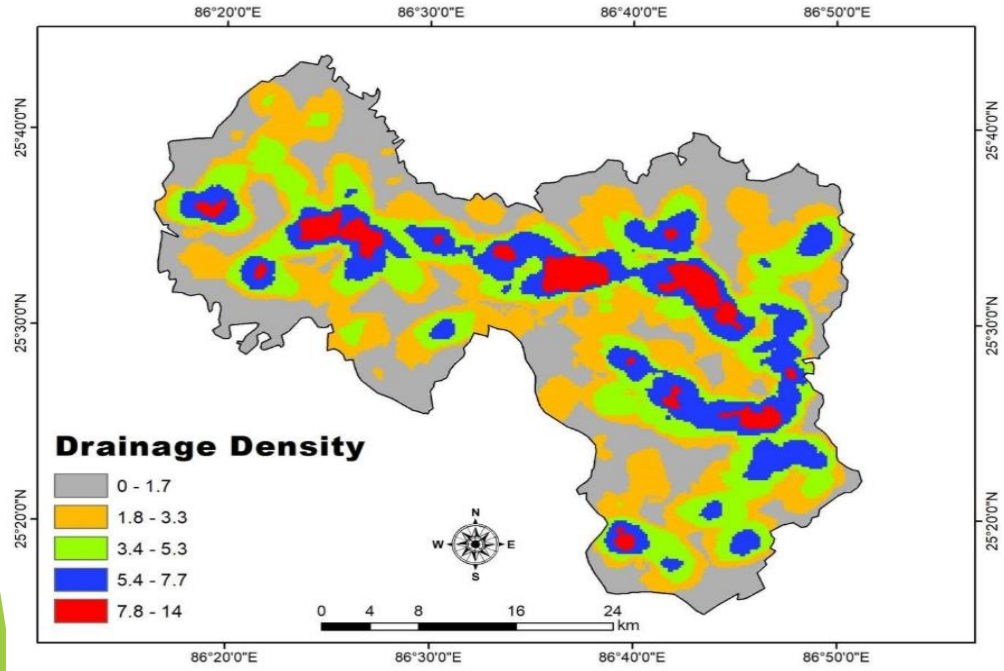


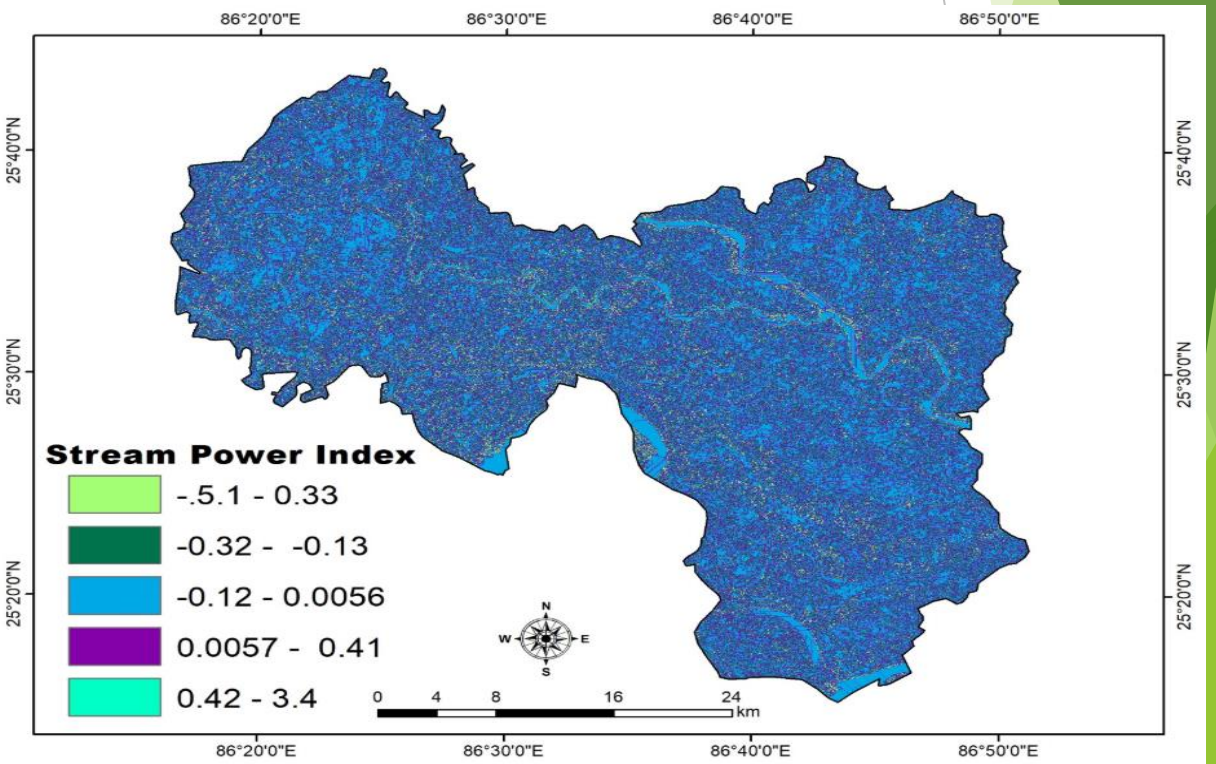


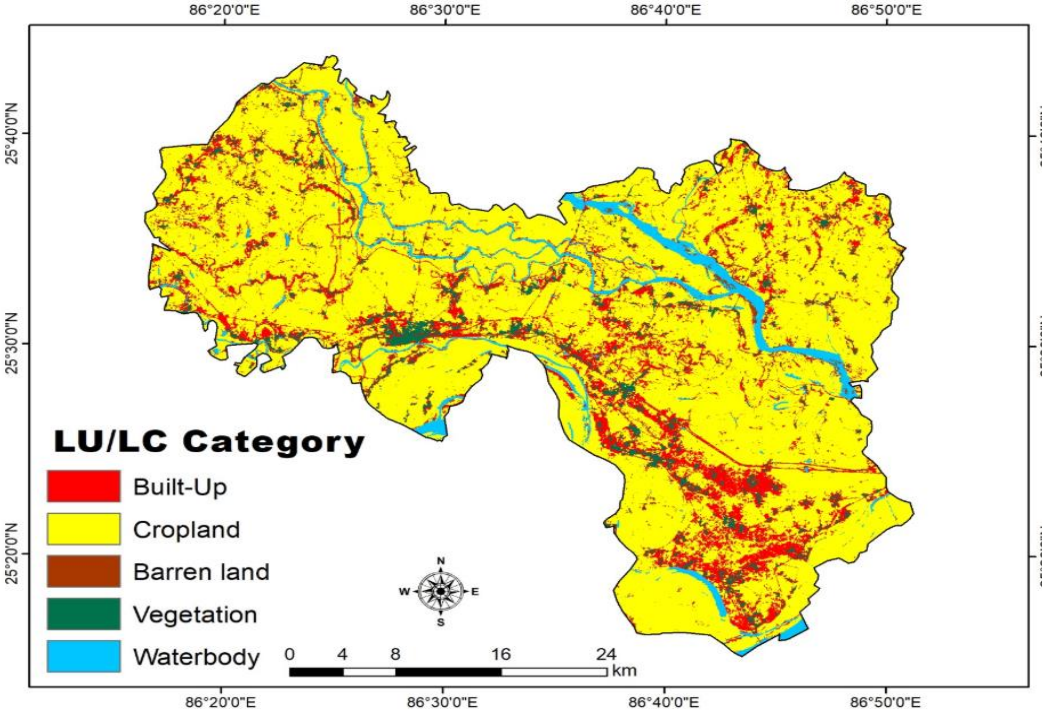
Maps of Various Indicators

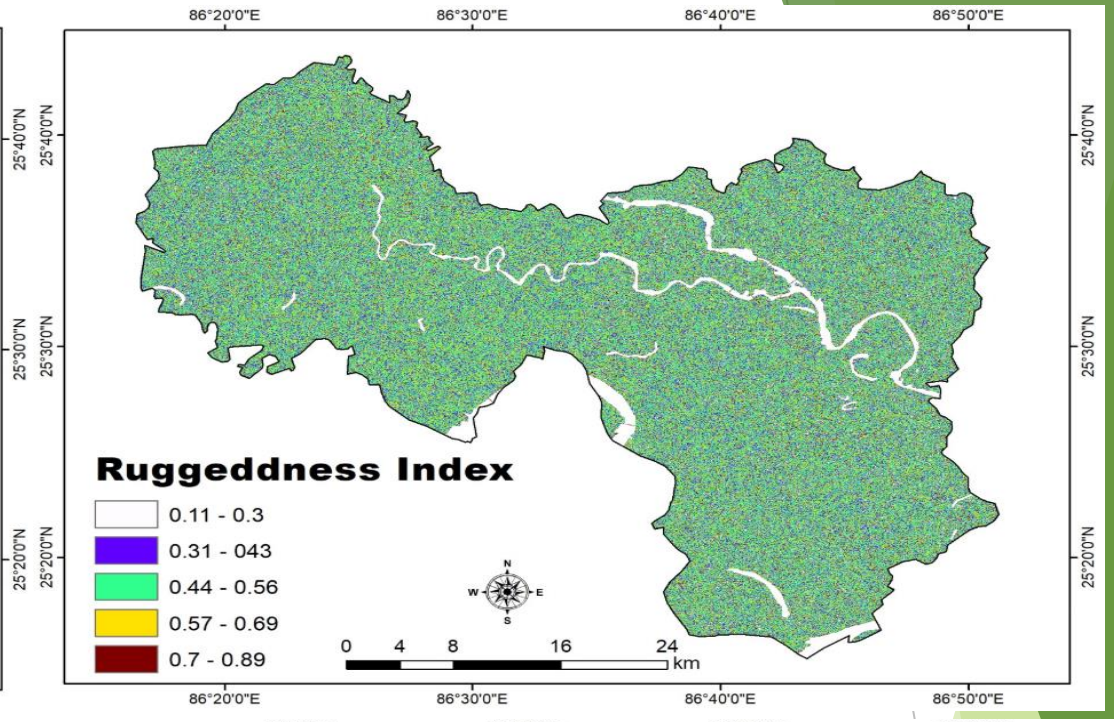

Distance from river (m)
0 - 500
501- 1000
1001- 2000
2001- 3000
above 3001
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N

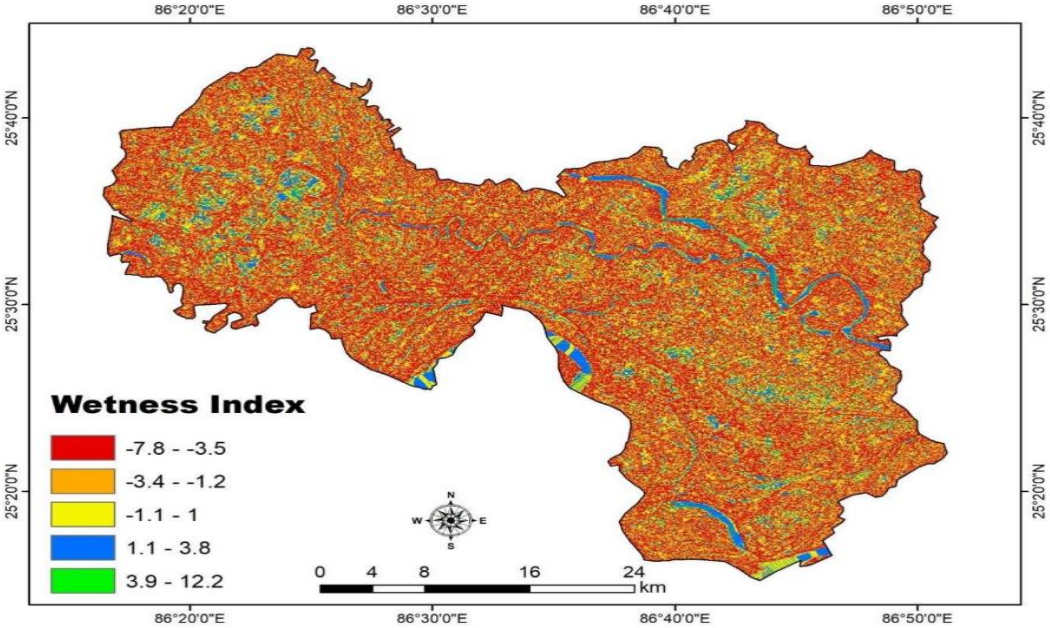

Rainfall (mm)
1,250.8 - 1,273.5
1,273.6 - 1,296.7
1,296.8 - 1,325.1
1,325.2 - 1,349.3
1,349.4 - 1,371.5
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N

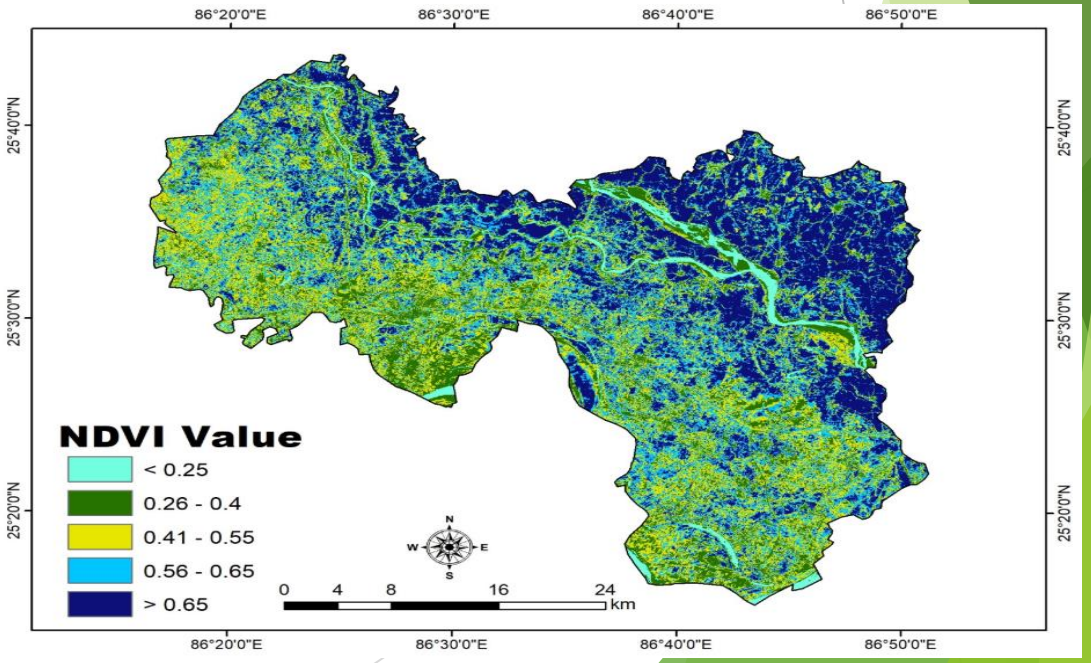

Drainage Density
0 - 1.7
1.8 - 3.3
3.4 - 5.3
5.4 - 7.7
7.8 - 14
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N


Stream Power Index
-.5.1 - 0.33
-0.32 - -0.13
-0.12 - 0.0056
0.0057 - 0.41
0.42 - 3.4
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N


LU/LC Category
Built-Up
Cropland
Barren land
Vegetation
Waterbody
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N


Ruggeddness Index
0.11 - 0.3
0.31 - 043
0.44 - 0.56
0.57 - 0.69
0.7 - 0.89
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N


Wetness Index
-7.8 - -3.5
-3.4 - -1.2
-1.1 - 1
1.1 - 3.8
3.9 - 12.2
0 4 8 16 24 km
86°20'0"E
86°30'0"E
86°40'0"E
86°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N


NDVI Value
< 0.25
0.26 - 0.4
0.41 - 0.55
0.56 - 0.65
> 0.65
0 4 8 16 24 km
86°20'0"E
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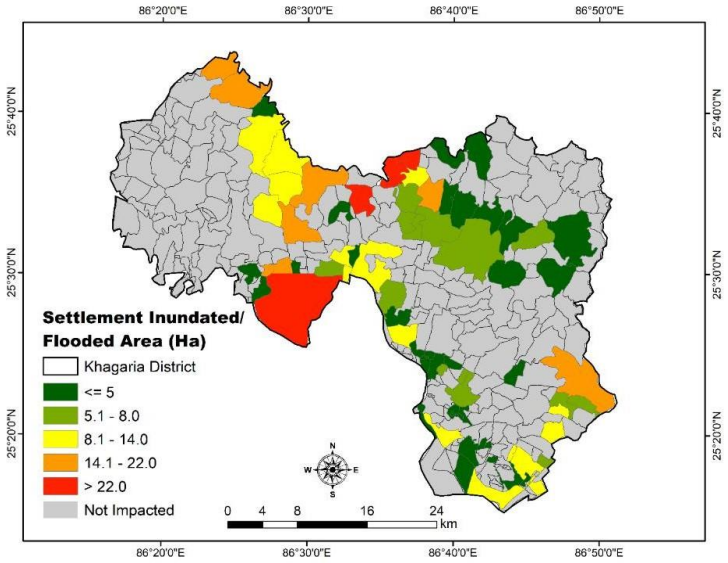
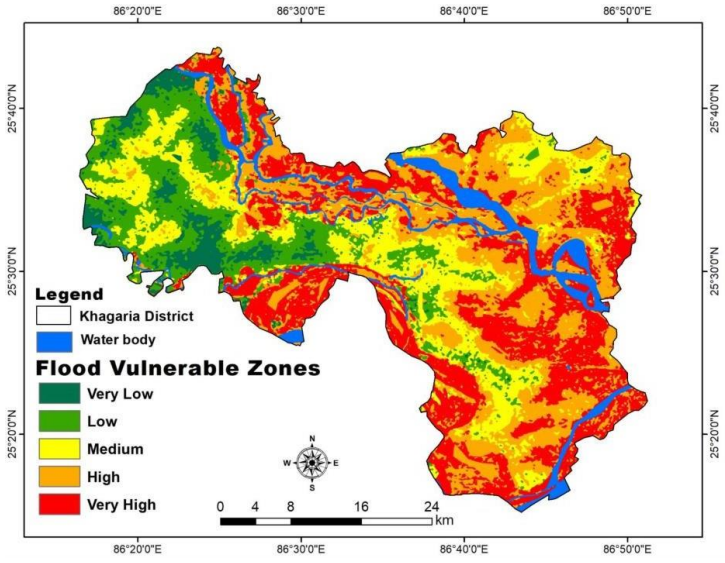
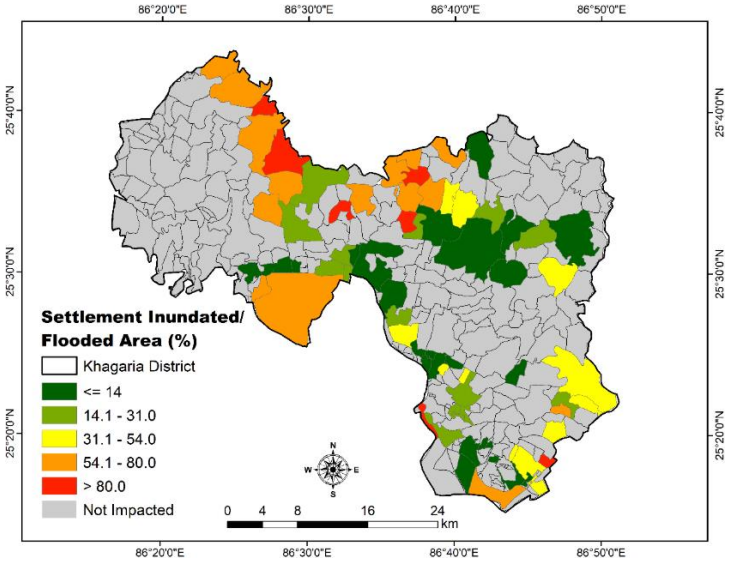
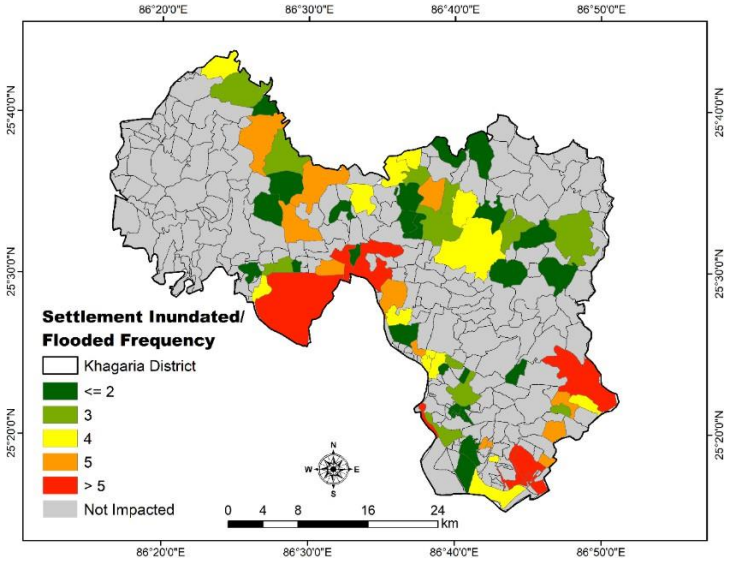
25°40'0"N
25°30'0"N
25°20'0"N

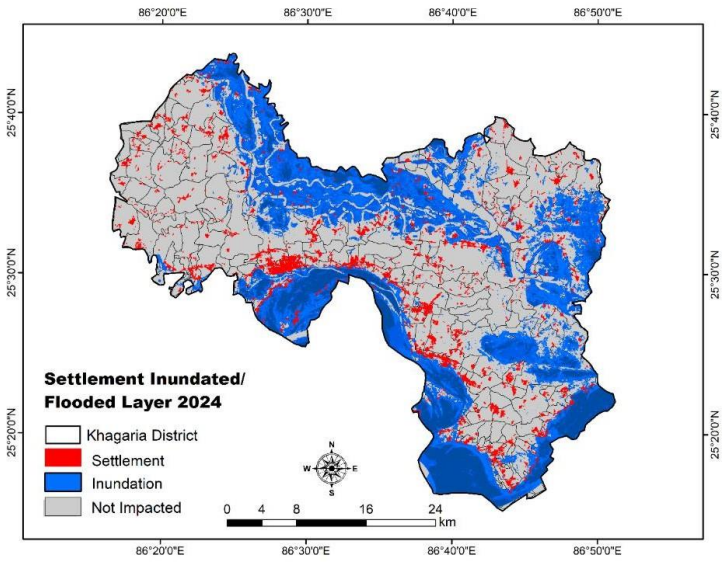


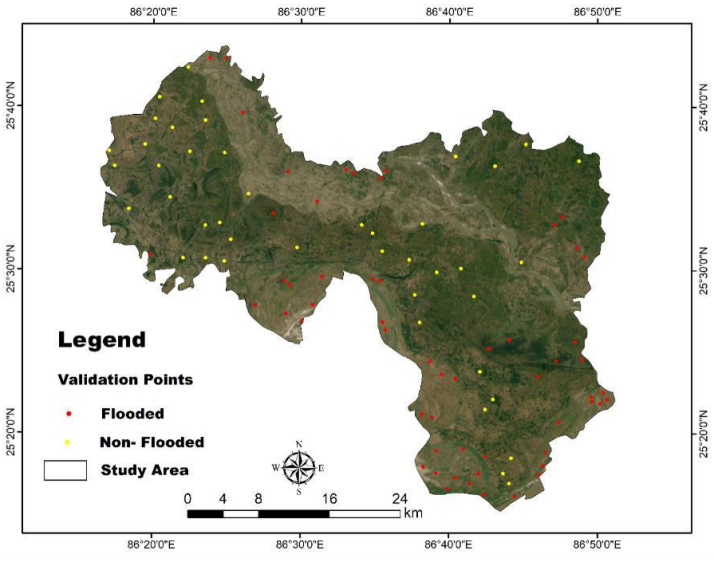




Flood Vulnerability and Settlement Vulnerability Maps

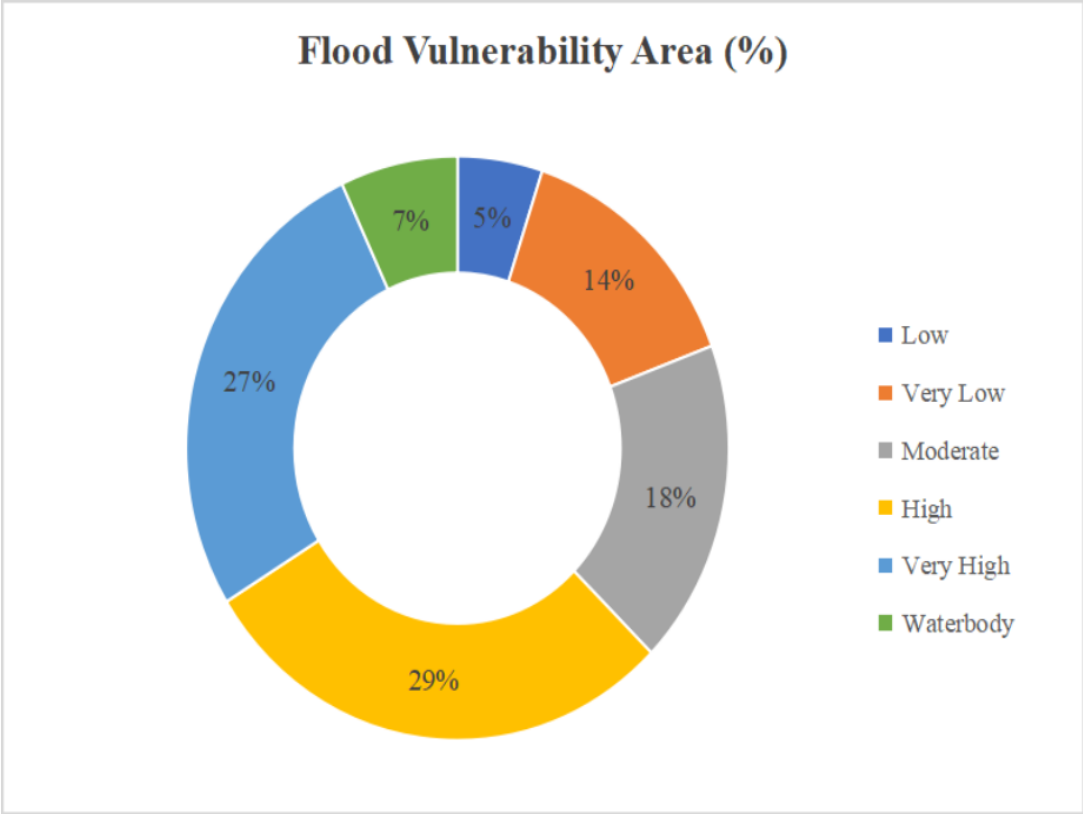




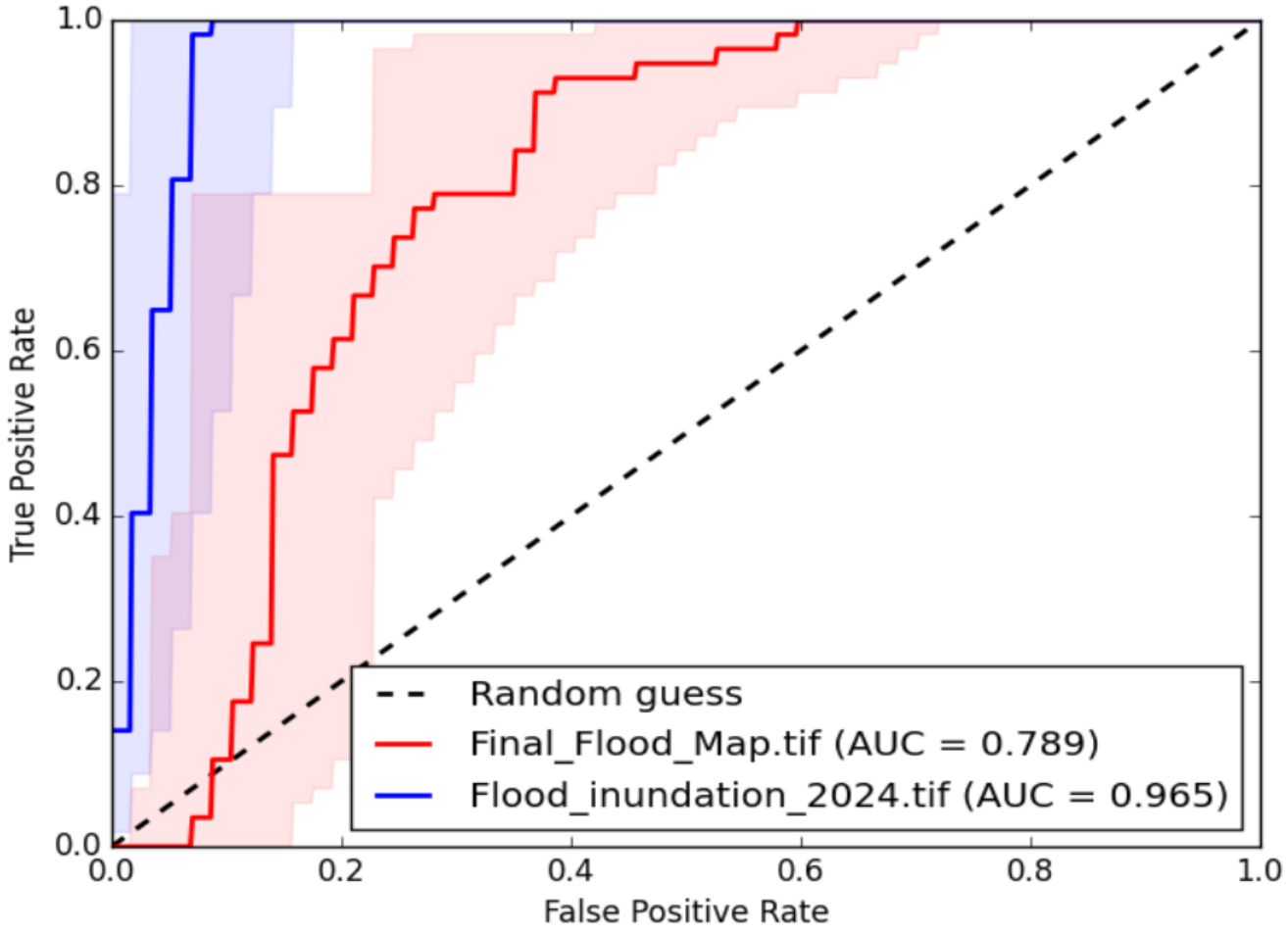


- Settlement Vulnerability
- And Flood Validation Map

Sr. No	Class	Area (Sq.km)
	Very Low	75.362
	Low	212.323
	Moderate	268.844
	High	429.909
	Very High	399.502
	Waterbody	104.221
	Total	1490.16



Flood vulnerable areas with their percentage



ROC and AUC curve

Conclusion

- **The study focusses on the Khagaria district, a highly flood-prone region in Bihar, which has been largely overlooked inprior geospatial flood research, thereby addressing a significant knowledge deficiency. The objective isto identify flood-prone areas in Khagaria district by utilising a multicriteria decision-making approach within ageographical information system, incorporating remotely sensed data. The final flood map was generated using theAHP method, which utilised an 11 × 11 decision matrix to denote the importance of the following parameters: elevation, slope, rainfall, DD, DFR, TWI, SPI, LULC, NDVI, lithology, and roughness index. Khagaria districtencompasses an area of 1,490.16 km², with 75.362 km² designated as very low flood-vulnerable, 212.323 km² as lowflood-vulnerable, 268.844 km² as moderately flood-vulnerable, 429.909 km² as highly flood-vulnerable, and 399.502km² as very highly flood-vulnerable.**
- **VillagesSuch as Imadpur, Dariapur, Siswa, Siadatpur Aguani Inglish, Siromanharpur, Shirinia, Duraja, Marachi, Katghara, Chandwa, Ragdhan Karari, Rampur, Durgapur, and Sathman are categorised as exhibiting extreme floodvulnerability due to their geographical area. The integrated 2024 flood inundation raster was utilised to determine thevillage-specific affected settlement areas (in hectares and percentage) and their inundation frequency, resulting in thepreparation of maps for the year 2024. The area under the curve (AUC) for the inundation layer is 0.968, while theAUC for the flood vulnerability zonation map is 0.789 which is acceptable. The results can substantially enhance localflood risk mitigation strategies, infrastructure advancement, land-use planning, and emergency preparednessinitiatives.**